

SOP-MAINT-007: Replacing Upper Punches — Rotary Tablet Press RTP-7

Department: Maintenance Revision: 01 Effective Date: 2024-01-15 Machine: Rotary Tablet Press — RTP-7, Line 3 Bay 7

Purpose

This procedure defines the standardized method for safe and GMP-compliant replacement of upper punches on the Rotary Tablet Press RTP-7. Proper replacement ensures consistent tablet weight, hardness, and prevents mechanical damage to turret tooling. This SOP also minimizes contamination risks and ensures traceability of tooling components.

Scope

This procedure applies to trained maintenance technicians and authorized engineering personnel responsible for preventive maintenance or corrective replacement of upper punches on RTP-7 machines. It is applicable during scheduled maintenance, product changeovers, or when punch wear or damage is detected.

Prerequisites

Tools Required

- Torque wrench (range 5–50 Nm, calibrated)
- Punch removal tool / extraction rod
- Feeler gauge set (0.02–1.0 mm)
- Lint-free cloths

- Nylon brush
- Isopropyl alcohol (IPA 70%)
- Flashlight or inspection lamp
- Allen key set (metric)
- Dial indicator (optional for alignment verification)

Safety Equipment

- Safety gloves (cut-resistant)
- Safety goggles
- Antistatic lab coat
- Safety shoes
- Dust mask or respirator (if powder residue present)

Before You Begin

- Ensure machine is stopped and isolated using Lockout-Tagout (LOTO) procedure.
- Confirm batch completion and line clearance as per GMP requirements.
- Obtain maintenance permit and supervisor approval.
- Ensure area is cleaned and free from product residue.
- Verify correct replacement punches with approved specifications and certificates.
- Record punch batch number in maintenance logbook.

■■ Safety Warnings

- Always perform Lockout-Tagout before accessing internal machine components.

- Never attempt punch removal while the turret is in motion.
- Avoid direct contact with sharp punch tips to prevent injury and contamination.
- Ensure no loose tools or parts are left inside the turret assembly.
- Do not use damaged or non-qualified punches as this can cause catastrophic machine failure.
- Maintain strict GMP hygiene to prevent cross-contamination.

Procedure

Step 1 — Machine Isolation and Verification

Shut down the RTP-7 machine using the standard shutdown sequence and isolate all power sources using Lockout-Tagout. Verify zero energy state by attempting a start command and confirming no movement occurs. Open the machine guards and visually inspect that the turret is completely stationary. Ensure all interlocks are disengaged only after proper authorization and document the isolation in the maintenance log.

Step 2 — Access the Turret Assembly

Carefully open the upper guard assembly and secure it in the open position. Use a flashlight to inspect the turret area for any residual powder or debris. Clean the turret surface using a lint-free cloth and IPA to prevent contamination during punch replacement. Ensure the work area is well-lit and organized before proceeding.

Step 3 — Identify and Mark Punch Positions

Rotate the turret manually using the handwheel until the first upper punch is accessible. Mark the punch position using a non-permanent marker or refer to turret indexing numbers. This ensures correct tracking and avoids misalignment during reassembly. Document any abnormal wear patterns observed on existing punches.

Step 4 — Loosen Punch Retention Mechanism

Using the appropriate Allen key, carefully loosen the punch retaining screws or caps. Apply steady torque and avoid excessive force to prevent thread damage. Typical loosening torque is approximately 15–20 Nm. Ensure the retaining mechanism is fully disengaged before attempting punch removal.

Step 5 — Remove the Upper Punch

Insert the punch removal tool into the punch bore and gently extract the upper punch vertically. Do not tilt or force the punch, as this may damage the guide bore. Inspect the removed punch for wear, scoring, or deformation. Place the removed punch in a designated tray labeled for rejected tooling.

Step 6 — Inspect Punch Bore and Guide

Thoroughly inspect the punch bore using a flashlight. Check for scratches, residue buildup, or misalignment. Clean the bore using a nylon brush and IPA. Use a feeler gauge to verify clearance; acceptable clearance should be within 0.02–0.05 mm. Any deviation beyond tolerance must be reported before proceeding.

Step 7 — Install New Upper Punch

Take the new qualified upper punch and verify its part number, dimensions, and certification. Lightly lubricate the punch barrel with approved pharmaceutical-grade lubricant. Insert the punch vertically into the bore, ensuring smooth entry without resistance. Do not force the punch; it should slide freely within tolerance.

Step 8 — Secure Punch and Apply Torque

Reinstall the retaining mechanism and tighten using a calibrated torque wrench. Apply a torque of 18–22 Nm as per OEM specification. Ensure uniform tightening across all punch positions to maintain turret balance. Double-check that the punch is firmly seated and properly aligned.

Step 9 — Repeat for All Punches

Rotate the turret sequentially and repeat Steps 4 through 8 for each upper punch. Maintain consistency in installation and torque application. Keep track of completed positions to avoid duplication or omission. Ensure all replaced punches belong to the same batch or approved combination.

Step 10 — Final Inspection and Cleaning

After all punches are replaced, perform a comprehensive inspection of the turret assembly. Remove any tools, debris, or cleaning materials. Wipe down surfaces with IPA and ensure no foreign particles remain. Close all guards and ensure proper locking.

Step 11 — Trial Run and Verification

Release Lockout-Tagout and restore power. Run the machine in manual mode at low speed (10–15 RPM) and observe punch movement. Check for abnormal noise, vibration, or binding. Gradually increase speed to operational levels and monitor compression force stability.

Step 12 — Documentation and Handover

Record all maintenance activities, including punch batch numbers, torque values, and observations in the maintenance logbook. Inform production and quality teams that the machine is ready for operation. Ensure QA clearance before starting production.

Acceptance Criteria

The machine must operate smoothly without abnormal vibration (≤ 5 mm/s). Compression force readings should remain stable within $\pm 5\%$ of set values. Visual inspection should confirm proper punch alignment and no oil leakage or residue. Trial tablets must meet weight and hardness specifications within defined limits.

Escalation

Stop the procedure immediately if punch insertion resistance is abnormal, bore damage is detected, or alignment issues persist. Escalate to the maintenance supervisor if torque values cannot be achieved or if repeated punch failures occur. Contact OEM support if turret damage or structural issues are suspected.

Related Documents

- SOP-MAINT-002: Lockout-Tagout Procedure
- SOP-QA-010: Line Clearance Procedure
- SOP-PROD-005: Tablet Compression Operation

Parts Required

- Upper Punch — TP-UP-02 — As required (typically full set of turret positions)
 - Punch Retaining Screws — PRS-11A — As required
 - Food-grade Lubricant — LUB-PH-01 — 50 ml per maintenance activity
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